

DecPort

Distinguishing Cross-Backbone Decision-Behavior Transfer from Frozen-Core Portability

A repository-grounded reconstruction of the experimental progression through the nonlinear final ship gate

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Research conducted with the support of artificial intelligence (AI).

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Decision 0007 preregistered protocol/implementation: d0aa96f4defc3ce202996be1f75d669b791d57fc

Decision 0006 accepted implementation: 0ab715530a3facf5d95cba9bedf6949f82ccdd5e

Abstract

DecPort investigated two related questions: whether input-specific probabilistic decision behavior can be transferred across heterogeneous frozen language-model backbones without target labels, and whether a particular learned frozen DecisionCore contributes reusable functionality after the backbone change. The study progressed from supervised shared-head transfer through label-free latent alignment and decision-space distillation to two accepted Open-Jev stages. Decision 0006 established reproducible, input-specific cross-backbone behavior distillation: correct-teacher distillation outperformed a decision-type- and width-matched mismatched-teacher control on both in-distribution (ID) and out-of-distribution (OOD) accuracy for SmolLM2, Gemma 3, and TinyLlama in all five seeds. However, its released Open-Jev core was a scalar Linear(2048->1) readout paired with an expressive adapter whose final trainable projection can absorb the readout direction, so the learned-core comparison could not identify reusable core-specific computation. Decision 0007 directly addressed that confound with a learned nonlinear DecisionCore, a deliberately weaker rank-128 linear target adapter, a layer-scale-matched random nonlinear core, mismatched-teacher and untrained controls, a target-specific distilled baseline, and preregistered SHIP/NO-SHIP criteria. In that final gate, learned-core minus matched-random-core accuracy gains were near zero or negative and no target passed the primary learned-core criterion; the learned shared core also did not provide practical value over the target-specific distilled baseline under the preregistered utility criterion. By contrast, correct-teacher versus mismatched-teacher transfer remained positive on all three targets in every seed, and the preregistered input-specific and OOD-transfer criteria passed. The combined evidence therefore supports label-free, input-specific decision-behavior distillation within the tested backbones, datasets, tasks, and training budget, while it does not support the original DecPort hypothesis that the learned shared DecisionCore itself provides useful reusable decision computation across those backbones.

Keywords: probabilistic decision models; knowledge distillation; model stitching; representation alignment; frozen backbones; cross-model transfer; calibration; reproducibility; preregistered falsification

1. Introduction

Probabilistic decision systems occupy a narrower interface than general text generation: given a state and a bounded decision specification, they return a typed choice or score together with a probability distribution. A portable decision layer would be operationally attractive if a learned decision capability could be separated from the language-model backbone that produced its representations. Such separation could permit the same learned decision module to be reused when the underlying backbone changes, while requiring only a small model-specific interface to the new representation space.

DecPort was constructed to test that proposition rather than assume it. Its core architectural hypothesis can be written as a family of frozen backbones B_i , trainable adapters A_i , and one shared frozen DecisionCore C : $B_i(x) \rightarrow A_i(B_i(x)) \rightarrow C(A_i(B_i(x)))$. The original research question was whether C could encode a learned decision capability whose functionality survived a change of backbone, rather than merely serving as an arbitrary fixed readout around which A_i could be trained. [P1-P6]

This distinction separates two claims that are easy to conflate. Behavior transfer asks whether a target backbone can reproduce useful, input-specific decision behavior from a source system without target ground-truth labels. DecisionCore portability is stronger: the particular learned frozen core must contribute reusable functionality across backbone changes, such that replacing it by an arbitrary matched frozen core materially degrades the transfer. A positive teacher-versus-mismatched-teacher result addresses the first claim. It does not, by itself, establish the second.

Accordingly, the paper evaluates two explicit research questions:

RQ1 - Behavior transfer. Can input-specific probabilistic decision behavior be transferred from a source decision system to a heterogeneous frozen target backbone without access to target ground-truth labels?

RQ2 - DecisionCore portability. Does the specific learned frozen DecisionCore contribute causally to successful transfer, beyond what an arbitrary matched frozen core can provide when the target adapter is trained under the same teacher signal?

RQ1 is operationalized by the correct-teacher versus mismatched-teacher contrast. RQ2 is operationalized by the learned-core versus matched random-core contrast. These comparisons are intentionally distinct: evidence for RQ1 does not imply

evidence for RQ2.

The repository records a sequence of increasingly restrictive experiments. Each stage was introduced because the previous stage admitted a simpler explanation: target-label supervision could train around an arbitrary head; latent matching could improve geometric alignment without preserving decisions; decision-space distillation could transfer source behavior without showing that the head itself was special; and scaling could demonstrate reproducibility without using a real external decision system. The sequence ultimately reaches two accepted Open-Jev stages. Decision 0006 used the released Open-Jev scalar linear core and established behavior transfer while exposing a structural identifiability problem. Decision 0007 was preregistered specifically to remove that explanation by replacing the linear core with a nonlinear learned DecisionCore and the expressive adapter with a constrained rank-128 linear map, while adding a scale-matched random nonlinear core and a target-specific baseline. [P6-P14, P20-P27]

2. Related Work

2.1 Knowledge distillation

Knowledge distillation transfers predictive information from a teacher into a student objective, classically by training the student to match the teacher’s softened output distribution rather than only hard labels [E1]. DecPort uses the same general output-matching principle in its decision-space stages, but the adaptation target is deliberately narrower: the target language-model backbone remains frozen, the downstream DecisionCore remains frozen, and only a lightweight backbone-specific adapter is optimized. Target ground-truth labels are excluded from the transfer objective. The resulting question is therefore not conventional model compression. It is whether the teacher’s input-specific decision function can be reconstructed across a change of representation system through a constrained interface.

2.2 Representation similarity and alignment

A separate line of work studies whether learned representations from different networks encode similar information. SVCCA compares representation subspaces through singular-value reduction followed by canonical correlation analysis [E9], while centered kernel alignment (CKA) was developed as a robust alternative for comparing neural representations across independently trained networks [E5]. Such measures characterize statistical or geometric correspondence; they do not by themselves establish that one representation can be substituted for another inside a frozen downstream computation. DecPort’s own latent-alignment diagnostics exposed the same distinction empirically: improved latent reconstruction did not monotonically translate into improved decision accuracy, and the alignment method with the lowest raw latent loss was not the method with the highest downstream accuracy [P16, P17].

2.3 Model stitching and functional interchangeability

Model stitching provides a more functional test of representational compatibility. Lenc and Vedaldi introduced learned transformation or stitching layers as a way to study equivalence between representations [E2]. Bansal, Nakkiran, and Barak later formalized stitching between frozen networks using a simple trainable interface and emphasized that functional stitchability can reveal relationships not captured by representation-similarity statistics alone [E3]. Hernandez et al. extended the methodology to representations of different shapes and networks with different architectures [E4]. DecPort is structurally related to this family: frozen upstream and downstream components are connected through a learned interface. Its causal question is different, however. The downstream component is a specific probabilistic DecisionCore, and portability requires more than successful stitching - the identity of that learned core must contribute functionality that an arbitrary matched frozen core cannot replace.

2.4 Positioning DecPort

DecPort can therefore be viewed as a constrained combination of behavior distillation and model stitching. RQ1 asks whether teacher behavior can be reconstructed through a lightweight adapter despite a backbone change. RQ2 asks whether successful reconstruction depends on the particular learned frozen core rather than merely on the existence of some fixed differentiable readout. The latter requirement is stronger than ordinary stitchability: if the adapter can absorb or reparameterize the downstream computation, high stitched performance does not identify reusable information in the frozen core itself. The experiments below are organized around progressively eliminating these alternative explanations. This positioning is methodological; it does not rely on a claim that no prior work has studied adjacent forms of cross-model transfer or representational compatibility.

3. Experimental progression

The progression is methodologically central because the strongest conclusion is not the endpoint alone, but the sequence by which alternative explanations were exposed and controlled. Unless explicitly marked otherwise, the experiments in Sections 3.1-3.6 are diagnostic evidence. They were retained to guide hypothesis refinement, not promoted to the accepted Open-Jev benchmark archives.

3.1 Supervised shared-head transfer

The initial design trained a source adapter and scalar decision head on labeled data, froze that head, and then trained a target adapter under the same labels. Both language-model backbones remained frozen. A fresh random frozen head served

as a control. The architecture appeared to realize the desired shape - different backbones mapped into a common latent space consumed by one head - but the target labels left an important identifiability gap. [P1, P15]

The diagnostic made that confound visible. On the five-seed bounded run, the shared-head advantage over a frozen random head was $+0.1000 \pm 0.0525$ after two epochs but narrowed to $+0.0063 \pm 0.0450$ after five epochs; final per-seed gains included one negative seed. The result suggested that the target adapter could learn around an arbitrary fixed head when given the labels needed to optimize target accuracy. The experiment therefore could not isolate portable head-specific information. [P15]

3.2 Random-head control as the first falsification pressure

The random-head comparison changed the research question. Supervised target performance was no longer a sufficient success criterion. A valid test had to deny the target the ground-truth answer during adaptation and require its behavior to be specified by the source system. That requirement motivated the label-free stages that followed. [P1, P2]

3.3 Label-free latent alignment

The next experiment froze the complete source system and the target backbone, removed target answers, and optimized only the target adapter to align target latent representations with source latent representations for the same state/question/candidate inputs. A cosine-plus-MSE objective reduced the alignment loss from 1.4863 to 0.7662 in the seed-0 diagnostic; ID accuracy improved from 0.3281 unaligned to 0.4219 aligned. The result was directionally positive but far below supervised/native controls and remained a single-seed bounded diagnostic. [P2, P16]

This stage eliminated direct target-label leakage but introduced a different question: does closeness in the chosen latent geometry imply preservation of the source’s decision function? The subsequent method comparison showed that it did not reliably do so.

3.4 Alternative alignment methods

DecPort compared cosine-plus-MSE alignment with source-covariance whitening, ridge affine regression, and centered orthogonal Procrustes. Ridge regression produced a lower latent reconstruction loss (0.6950 in the archived diagnostic) but lower ID decision accuracy (0.4062) than cosine-plus-MSE (0.4219); Procrustes also reached 0.4062, while whitening reached 0.2969. These diagnostics showed that a representation-level objective could improve geometric reconstruction without monotonically improving the actual decision behavior read by the frozen head. [P3, P17]

3.5 Label-free decision-space distillation

The next step moved the adaptation objective from latent coordinates to the teacher’s own decision logits. Teacher outputs were cached without answers; the target adapter alone was trained through the frozen source head using a temperature-scaled KL term plus centered-logit MSE. The control permuted teacher outputs within compatible option-count groups, preserving marginal output structure while destroying the input-to-output association. This control directly tests whether the adapter is learning input-specific teacher behavior rather than only generic score statistics. [P4, P18]

Across three diagnostic seeds, decision-space distillation reached 0.6615 ± 0.0549 ID accuracy versus 0.3438 ± 0.0563 unaligned, 0.3802 ± 0.0477 latent alignment, and 0.3229 ± 0.0180 under the permuted teacher. Correct-teacher minus permuted-teacher accuracy was $+0.3385 \pm 0.0705$. Teacher-student top-choice agreement reached 0.8438 ± 0.0625 and probability correlation 0.8372 ± 0.0652 on the distillation inputs. The BoolQ OOD gain was much smaller and accompanied by worse probabilistic loss and calibration, so the run was interpreted as bounded positive evidence for behavior transfer, not general portability. [P18]

3.6 Scaled multi-backbone decision transfer

A five-seed scale diagnostic increased the earlier splits by tenfold to 1,280 train, 640 ID, and 320 BoolQ OOD examples and added Gemma alongside SmoLM2. With a frozen Qwen source head, correct-teacher distillation reached 0.7828 ± 0.0173 ID accuracy on SmoLM2 and 0.7888 ± 0.0295 on Gemma, versus 0.3713 ± 0.0130 and 0.3813 ± 0.0301 respectively under the permuted teacher. The effect held in every seed and teacher-student probability correlation exceeded 0.85 on ID evaluation for both targets. OOD gains remained small and noisy and calibration worsened. [P5, P19]

This stage strengthened the behavior-transfer result across target families, but the source head was still a DecPort-trained Qwen head. An accepted experiment was needed to test the architecture against an external, independently released decision system and to include the control that directly asks whether the specific pretrained core matters.

3.7 First accepted Open-Jev stage: linear DecisionCore (decision 0006)

The first accepted Open-Jev experiment imported Open-Jev’s released 2B model package and its trained scalar decision head, pinned every relevant model and code revision, expanded evaluation across Choice, Noul, and Score decisions, used three heterogeneous frozen target backbones, and ran five seeds. Most importantly, it compared the learned Open-Jev core with a random frozen core matched in weight norm, bias, and calibration temperature. The correct-versus-mismatched contrast established input-specific behavior transfer, while the learned-versus-random comparison exposed a structural confound: the scalar linear core can be absorbed or reparameterized by the expressive adapter. Decision 0006 therefore remains valid evidence, but it is evidence that motivates a stronger RQ2 test rather than the final test itself. [P6-P14]

3.8 Final nonlinear ship gate (decision 0007)

Decision 0007 was designed specifically around the 0006 identifiability result. It learned a nonlinear DecisionCore once on source-only Open-Jev representations, froze it, reduced the target adapter to a rank-128 linear map with no activation, constructed a scale-matched random core with the same nonlinear architecture, retained the mismatched-teacher and untrained controls, and added a target-specific distilled module. Source-core training records were disjoint from all transfer and evaluation records so that target adaptation remained label-free. The architecture, data, training budget, control construction, and SHIP/NO-SHIP thresholds were frozen at commit d0aa96f4defc3ce202996be1f75d669b791d57fc before the full run. The accepted result archive was committed at cf9939e213ca2c045bc6bf6968b68bcc8f8656be. This became the culminating falsification test of RQ2. [P20-P27]

4. Open-Jev linear-core experiment (0006): experimental design

The accepted archive is `benchmarks/accepted/openjev-transfer-v0.1/2026-09-23-wsl2-rtx4060ti-seeds0-4/` at DecPort evidence commit 51f221348163b320e46111d0857a4a255e304ec0. The experiment implementation recorded in provenance is DecPort commit 0ab715530a3facf5d95cba9bedf6949f82ccdd5e. [P7-P10]

4.1 Teacher, core boundary, and frozen components

The teacher is ZefanCai/Open-Jev-2B at model revision 0c7aa498b1627be8da4acf34c863ff0ee0a92785, loaded with the pinned Open-Jev code at commit 3308a15ccd7eeald7a37d6ddc39b023b801ba16. The package records Qwen/Qwen3.5-2B revision 15852e8c16360a2fea060d615a32b45270f8a8fc, LoRA rank 8, maximum package length 4096, and saved calibration temperature 1.518796342858676. Open-Jev itself uses one shared scalar candidate scorer and typed assembly for Choice, Noul, and Score decisions. [P10, E6, E7]

DecPort defined the DecisionCore boundary as the released Open-Jev float32 Linear(2048->1) head, its calibration temperature, and typed assembly semantics. Choice and Score obtain one scalar per candidate and softmax across candidates. Noul obtains a single scalar s and assembles logits $[0, s]$ for (false, true), so $P(\text{true}) = \sigma(s/T)$. The core state SHA-256 in the accepted run is 867bd23f8a8892fcb411dda861b1d5d53d74dea86d73b9064d9057ccce973f0. [P7, P10, P14]

The Open-Jev Qwen backbone, its LoRA adapter, the decision head, the calibration temperature, every target backbone, and the random control core were frozen. Only each target’s DecPort adapter was optimized. Source teacher outputs were detached, target distillation records contained no answers, and the transfer entry point rejected labeled records. [P11, P14]

4.2 Target backbones and adapters

The three frozen targets were:

- HuggingFaceTB/SmolLM2-360M-Instruct, revision a10cc1512eabd3dde888204e902eca88bddb4951;
- unsloth/gemma-3-270m-it, revision 23cf460f6bb16954176b3ddcc8d4f250501458a9;
- TinyLlama/TinyLlama-1.1B-Chat-v1.0, revision fe8a4ea1ffedaf415f4da2f062534de366a451e6.

[P10]

Each trainable adapter had the same functional form, parameterized by target hidden width d :

$$\text{LayerNorm}(d) \rightarrow \text{Linear}(d, 256) \rightarrow \text{GELU} \rightarrow \text{Linear}(256, 2048).$$

Thus the adapter was capable of a nonlinear remapping before a final linear projection into the 2,048-dimensional space expected by the frozen DecisionCore. All four conditions for a given target and seed began from exactly the same adapter initialization. [P14]

4.3 Data, splits, and objective

The archived jev-broad-v0.1 data contain 4,500 transfer-training decisions: 1,500 Choice from ARC-Easy, 1,500 Noul/Boolean from BoolQ, and 1,500 Score from Yelp Review Full. The ID evaluation has 2,000 labeled decisions (570 Choice, 700 Noul, 730 Score) from the corresponding held-out source datasets. The OOD evaluation has 1,500 labeled decisions, 500 per type, from OpenBookQA (Choice), QNLI (Noul), and Amazon Reviews Multi (Score). The data manifest reports zero exact-record overlap across splits and records dataset fingerprints and SHA-256 digests for each generated split. [P9]

Training used seeds 0-4, ten epochs, AdamW with learning rate 10^{-3} and weight decay 0.01, gradient clipping at 1.0, batch size 8, target maximum length 2,048, and teacher maximum length 4,096. No target-label early stopping, selection, weighting, or tuning was used. [P9-P11]

For calibrated typed logits ℓ_t from the teacher and ℓ_s from the student, the target-adapter objective was

$$L = \text{KL}(p_{\text{teacher}} \parallel p_{\text{student}}) + 0.1 \text{MSE}(\text{center}(\ell_s), \text{center}(\ell_t)), \quad T = 1.$$

The centered-logit term retains relative score geometry without penalizing a common additive offset. Ground-truth target labels were used only after training for evaluation metrics. [P7, P9, P14]

4.4 Four target conditions and their causal purpose

- **Untrained adapter.** Measures the behavior of the matched initial adapter through the learned Open-Jev core before adaptation.

- **Correct Open-Jev teacher distillation.** Trains the adapter against the teacher output produced for the same input. This is the primary behavior-transfer condition.
- **Mismatched-teacher distillation.** Deranges teacher outputs within decision kind and option count so that marginal distributional structure is retained while the input-specific association is destroyed. One singleton five-option Choice record cannot be deranged and is explicitly retained as a fixed point (1/4,500 per seed). This makes the control slightly closer to the correct teacher and is therefore conservative with respect to the behavior-transfer hypothesis.
- **Random frozen core.** Replaces the learned Open-Jev readout direction by a seeded random direction normalized to the learned weight norm while preserving the learned bias and calibration temperature; the adapter is then trained against the correct teacher outputs. This asks whether the particular learned readout is needed, rather than whether an adapter can learn through some fixed scalar readout.

The correct-versus-mismatched contrast identifies input-specific behavior transfer. The learned-core-versus-random-core contrast is the central 0006 test of specific DecisionCore portability. [P7, P14]

4.5 Statistical treatment

All uncertainty reported for the accepted internal experiment treats the random seed, not the individual decision, as the replication unit. Means and standard deviations are computed across the five paired seeds using the sample standard deviation ($n - 1$). For the primary RQ2 comparison, the learned-core and random-core conditions share the same seed, target backbone, teacher outputs, and matched adapter initialization; the analysis therefore uses the paired per-seed accuracy difference $d = \text{accuracy}_{\text{learned}} - \text{accuracy}_{\text{random}}$. Table C reports a two-sided 95% Student- t confidence interval for the mean paired difference with $n = 5$ and 4 degrees of freedom:

$$\text{CI}_{95} = \text{mean}(d) \pm t_{0.975,4} \frac{s_d}{\sqrt{5}}.$$

This confidence-interval analysis is post hoc and descriptive; no confirmatory hypothesis test or equivalence margin was preregistered for 0006. With only five seeds, the intervals are necessarily imprecise. In particular, an interval containing zero is evidence that 0006 does not resolve a stable learned-core advantage at this precision; it is not proof that the two cores are equivalent.

5. Integrity and reproducibility of decision 0006

The accepted archive was treated as a measurement package rather than a collection of selected headline metrics. It includes repository and expanded configurations, a data manifest, exact model/software provenance, per-seed results, retained model artifacts, an aggregate summary, rendered tables, an audit report, a second execution, and archive-wide SHA-256 checksums. [P7-P12]

The audit verifies strict target-label exclusion: all 4,500 distillation decisions per seed have answer=None; teacher inputs contain no answers; target-label early stopping or tuning is disabled; and a dedicated guard rejects labeled transfer records. It also verifies the gradient boundary on every trained condition: all adapter tensors receive gradients while no DecisionCore or backbone tensor does. The learned Open-Jev core digest is checked before and after transfer and against an independently reloaded copy from the pinned package. Teacher-versus-DecPort core replay has maximum absolute logit difference 0.0. [P11, P14]

Every-decision accounting is explicit. The audit checks per-type and per-dataset counts against the manifest and rejects missing decisions. The mismatched-teacher control is checked to have exactly the one unavoidable singleton fixed point. A tampered copy was used to exercise failure paths: an injected core gradient and a dropped Noul decision were both rejected. [P7, P11]

The reproduction directory records an independent re-execution of the identical command, code, data, software stack, and RTX 4060 Ti hardware environment. The two executions had maximum numerical difference 0.0 across all compared result JSONs; all 70 result JSON files compared numerically identically; aggregate_summary.json was byte-identical; all 75 retained safetensors artifacts were byte-identical; and the cached teacher logits were byte-identical. [P12]

This is strong deterministic repeatability evidence for the archived implementation. It is not an independent-laboratory replication, and it is not a cross-hardware or cross-software reproducibility study. Both runs used the same hardware/software setting. That boundary is retained in the interpretation below.

6. Results of decision 0006

Unless stated otherwise, accepted internal results are mean $\pm$ sample standard deviation across five seeds. ID accuracy uses $N = 2,000$ decisions per seed; OOD accuracy uses $N = 1,500$ decisions per seed. The Open-Jev teacher is deterministic for the archived pass, so its reference accuracies are not seed-averaged. For the paired learned-core versus random-core comparison, Table C additionally reports two-sided 95% Student- t confidence intervals over the five paired seed differences. [P7-P9]

Table A. Decision 0006 overall ID/OOD accuracy by target and condition

Target	Split	Correct teacher	Mismatched	Untrained	Random core
SmolLM2	ID	0.4602 ± 0.0074	0.2823 ± 0.0281	0.3176 ± 0.0449	0.4588 ± 0.0082
SmolLM2	OOD	0.3755 ± 0.0093	0.3112 ± 0.0200	0.3219 ± 0.0119	0.3716 ± 0.0047
Gemma 3	ID	0.4659 ± 0.0284	0.2806 ± 0.0249	0.3202 ± 0.0396	0.4370 ± 0.0119
Gemma 3	OOD	0.3799 ± 0.0135	0.3115 ± 0.0160	0.3268 ± 0.0105	0.3672 ± 0.0091
TinyLlama	ID	0.4478 ± 0.0023	0.2727 ± 0.0140	0.3190 ± 0.0311	0.4453 ± 0.0029
TinyLlama	OOD	0.3929 ± 0.0048	0.3208 ± 0.0049	0.3044 ± 0.0095	0.3875 ± 0.0085

Source: DecPort commit 51f2213; accepted 0006 archive `transfer/aggregate_summary.json` and `transfer/report_tables.md`. $n = 5$ seeds; SD is sample SD ($n - 1$). Teacher reference accuracy is 0.7310 ID and 0.6467 OOD.

Table B. Decision 0006 correct-teacher minus mismatched-teacher accuracy effect

Target	ID effect	ID seeds > mismatch	OOD effect	OOD seeds > mismatch
SmolLM2	$+0.1779 \pm 0.0266$	5/5	$+0.0643 \pm 0.0206$	5/5
Gemma 3	$+0.1853 \pm 0.0284$	5/5	$+0.0684 \pm 0.0086$	5/5
TinyLlama	$+0.1751 \pm 0.0141$	5/5	$+0.0721 \pm 0.0024$	5/5

Source: DecPort commit 51f2213; accepted 0006 archive. Effects are computed per seed before aggregation; $n = 5$; sample SD.

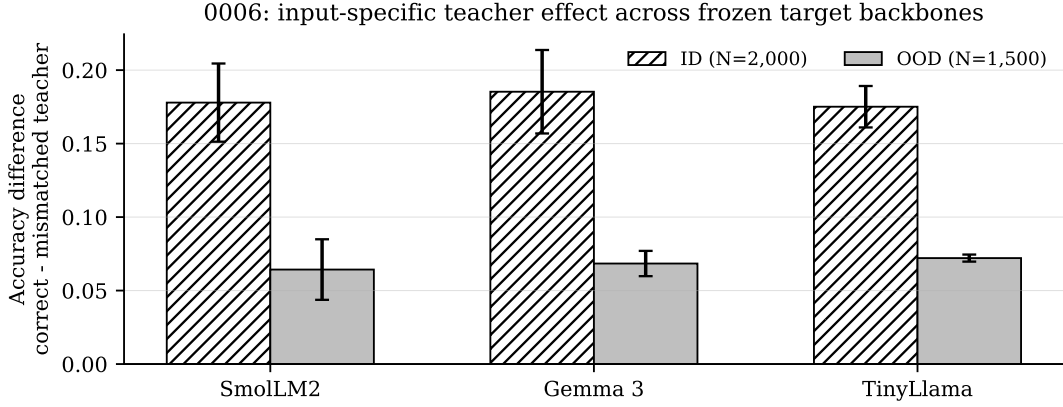


Figure 1: Decision 0006 input-specific teacher effect. Same accepted values as Table B. Error bars are sample SD across five seeds, not confidence intervals.

Table C. Decision 0006 learned Open-Jev core minus matched random frozen core

Target	ID Δ acc.	ID 95% CI	OOD Δ acc.	OOD 95% CI	Seeds > random ID/OOD
SmolLM2	$+0.0014 \pm 0.0112$	$[-0.0125, +0.0153]$	$+0.0039 \pm 0.0054$	$[-0.0028, +0.0106]$	3/5 / 4/5
Gemma 3	$+0.0289 \pm 0.0319$	$[-0.0107, +0.0685]$	$+0.0127 \pm 0.0202$	$[-0.0124, +0.0378]$	4/5 / 3/5
TinyLlama	$+0.0025 \pm 0.0033$	$[-0.0016, +0.0066]$	$+0.0055 \pm 0.0060$	$[-0.0019, +0.0129]$	4/5 / 5/5

Source: Paired by seed; $n = 5$; sample SD. 95% intervals are post hoc Student- t intervals with $df=4$. The random linear cores preserve learned-core weight norm, bias, and temperature while replacing readout direction.

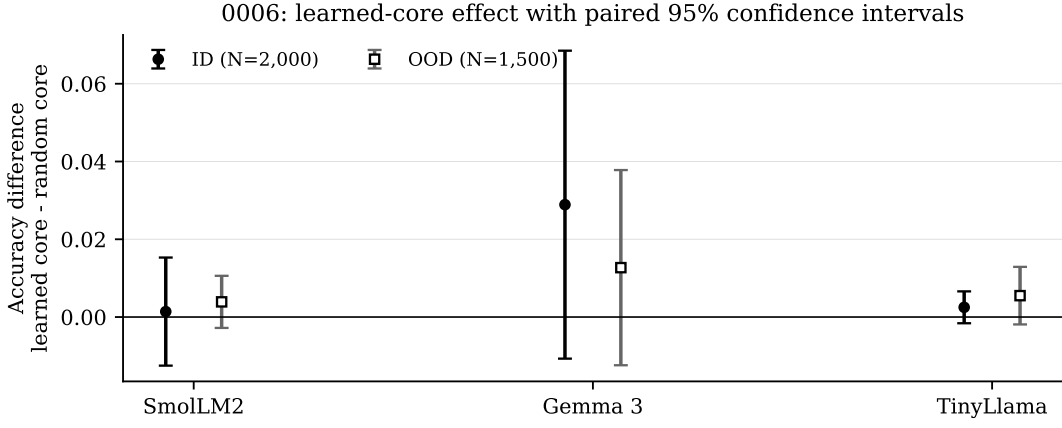


Figure 2: Decision 0006 learned-core minus random-core accuracy. Error bars are the paired 95% Student-*t* confidence intervals in Table C. All six intervals include zero; this is not an equivalence test.

Table D. Decision 0006 accuracy by decision type

Target	Split	Type (N)	Teacher	Correct	Mismatch	Random core
SmolLM2	ID	Choice (570)	0.8877	0.6446 ± .0118	0.2253 ± .0588	0.6414 ± .0149
SmolLM2	ID	Noul (700)	0.7929	0.3631 ± .0056	0.3849 ± .0532	0.3680 ± .0140
SmolLM2	ID	Score (730)	0.5493	0.4093 ± .0145	0.2285 ± .0121	0.4033 ± .0087
SmolLM2	OOD	Choice (500)	0.6080	0.3136 ± .0111	0.2296 ± .0271	0.3080 ± .0097
SmolLM2	OOD	Noul (500)	0.8600	0.5176 ± .0132	0.5096 ± .0311	0.5240 ± .0000
SmolLM2	OOD	Score (500)	0.4720	0.2952 ± .0186	0.1944 ± .0065	0.2828 ± .0084
Gemma 3	ID	Choice (570)	0.8877	0.4596 ± .0135	0.2330 ± .0426	0.4389 ± .0170
Gemma 3	ID	Noul (700)	0.7929	0.5066 ± .0669	0.3954 ± .0347	0.5020 ± .0199
Gemma 3	ID	Score (730)	0.5493	0.4318 ± .0176	0.2077 ± .0206	0.3732 ± .0205
Gemma 3	OOD	Choice (500)	0.6080	0.2580 ± .0147	0.2608 ± .0296	0.2568 ± .0044
Gemma 3	OOD	Noul (500)	0.8600	0.5340 ± .0224	0.4796 ± .0387	0.5260 ± .0032
Gemma 3	OOD	Score (500)	0.4720	0.3476 ± .0293	0.1940 ± .0079	0.3188 ± .0287
TinyLlama	ID	Choice (570)	0.8877	0.4881 ± .0052	0.2463 ± .0313	0.4807 ± .0135
TinyLlama	ID	Noul (700)	0.7929	0.3606 ± .0022	0.3617 ± .0012	0.3614 ± .0000
TinyLlama	ID	Score (730)	0.5493	0.5000 ± .0078	0.2079 ± .0156	0.4981 ± .0076
TinyLlama	OOD	Choice (500)	0.6080	0.3028 ± .0095	0.2500 ± .0206	0.3020 ± .0117
TinyLlama	OOD	Noul (500)	0.8600	0.5052 ± .0076	0.5236 ± .0009	0.4948 ± .0198
TinyLlama	OOD	Score (500)	0.4720	0.3708 ± .0064	0.1888 ± .0098	0.3656 ± .0129

Source: DecPort commit 51f2213; accepted 0006 archive. Target conditions are mean ± sample SD over five seeds; teacher is one deterministic archived pass.

7. What transferred in decision 0006

The strongest positive 0006 result is specific and reproducible: lightweight target adapters learned input-specific decision behavior from a real external Open-Jev system across three heterogeneous frozen language-model backbones without access to target ground-truth labels. The evidence for input specificity is the mismatched-teacher control, not target accuracy alone. Correct-teacher distillation exceeded that control on overall ID and OOD accuracy in all five seeds for every target (Table B), with ID effects of approximately +0.18 and OOD effects of approximately +0.06 to +0.07. [P7, P8]

The per-type decomposition shows where that effect originates. Choice transfer was strong in distribution for all three targets: correct-minus-mismatch accuracy was +0.4193 on SmolLM2, +0.2267 on Gemma, and +0.2418 on TinyLlama. Choice OOD retained positive mean effects for SmolLM2 (+0.0840) and TinyLlama (+0.0528) but not Gemma (-0.0028). Thus Choice behavior transferred strongly on the source-domain evaluation but did not generalize uniformly across the OOD dataset shift. [P7, P8]

Score was the most consistent transfer type. Correct-teacher distillation exceeded the mismatched teacher for all three targets on both ID and OOD accuracy, and this ordering held in 5/5 seeds per target in the archived analysis. Ordinal error also improved substantially: for example, TinyLlama’s ID ordinal MAE was 0.7049 ± 0.0321 under the correct teacher versus 1.8710 ± 0.2755 under mismatch; the corresponding OOD values were 1.0108 ± 0.0323 versus 1.8896 ± 0.2903 . The teacher reference was 0.574 ID and 0.752 OOD. [P7, P8]

Behavioral matching metrics support the same reading. The accepted archive reports overall teacher top-choice agreement for correct-teacher targets around 0.52-0.53 ID and 0.43-0.48 OOD, and correct-teacher KL is lower than untrained or mismatched conditions in the relevant transferred types. These values do not imply that the student recreates the teacher completely; target accuracies remain substantially below the Open-Jev teacher. They do show that the mapping learned from the teacher is input-dependent rather than a marginal or constant-output artifact. [P7, P8]

Accordingly, the positive 0006 result is best described as adapter-mediated, cross-backbone behavior distillation. It is stronger

than a statement that the adapters merely improved accuracy, because the correct-versus-mismatched contrast isolates the association between each input and its teacher output. It is weaker than a claim that the same learned computational core was functionally reused, because that latter claim remained unidentified by the linear-core experiment discussed in Section 9.

8. What did not transfer in decision 0006: Noul

Noul must be analyzed separately because aggregate accuracy hides its failure mode. The teacher’s Noul accuracy was 0.7929 ID and 0.8600 OOD. SmolLM2’s correct-teacher Noul accuracy was only 0.3631 ± 0.0056 ID, below both its untrained mean (0.4726 ± 0.1515) and its mismatched-teacher mean (0.3849 ± 0.0532). TinyLlama was similarly 0.3606 ± 0.0022 ID, essentially identical to mismatch (0.3617 ± 0.0012) and random core (0.3614 ± 0.0000). Gemma showed a limited positive signal at 0.5066 ± 0.0669 ID versus 0.3954 ± 0.0347 under mismatch, but it remained far below the teacher and essentially matched its random-core condition (0.5020 ± 0.0199). [P7, P8]

The collapse is visible on the transfer-training inputs themselves. For Noul, teacher-student probability correlation on the training decisions was only 0.063 ± 0.010 for SmolLM2 and 0.048 ± 0.024 for TinyLlama. Their Noul KL barely changed from the untrained adapter: 0.2885 to 0.2858 for SmolLM2 and 0.2869 to 0.2867 for TinyLlama. Gemma reached a higher but still limited training correlation of 0.300 ± 0.104 . [P7]

Predicted-class behavior explains why superficially nonzero Noul accuracy is misleading. SmolLM2 and TinyLlama predict true for only about 1% of ID Noul decisions. Their approximately 0.36 ID accuracy therefore matches the false-label prevalence (about $1 - 0.6386 = 0.3614$) rather than indicating recovery of input-specific teacher behavior. The archived analysis notes that the students approximately reproduce the teacher’s marginal $P(\text{true})$ near 0.48 while failing to reproduce which inputs should be assigned high or low probability. This is a marginal-behavior collapse, not simply an OOD generalization failure, because it is already present on the training decisions. [P7]

OOD Noul performance is correspondingly near chance/majority behavior: correct-teacher accuracy is 0.5176 ± 0.0132 for SmolLM2, 0.5340 ± 0.0224 for Gemma, and 0.5052 ± 0.0076 for TinyLlama, versus a QNLI majority-class rate of 0.524 and teacher accuracy of 0.8600. TinyLlama seed 0 additionally diverged on QNLI with mean $P(\text{true})$ about 0.92 and Noul KL 2.43, producing a large seed-specific OOD calibration error. [P7, P8]

The Noul result constrains the positive conclusion. Decision 0006 did not transfer all supported decision semantics uniformly. The aggregate behavior-transfer effect is driven by Choice and especially Score, and any summary that omitted the Noul decomposition would materially overstate the breadth of the result.

9. Decision 0006 random-core result and the identifiability problem

The random-core control is the scientifically decisive 0006 result for RQ2. Across all three targets, replacing the trained Open-Jev readout by a norm-, bias-, and temperature-matched random frozen readout caused only small accuracy changes after the adapter was trained against the same correct teacher outputs. The learned-minus-random differences are shown in Table C and Figure 2. Every paired 95% t confidence interval includes zero: SmolLM2 $[-0.0125, +0.0153]$ ID and $[-0.0028, +0.0106]$ OOD; Gemma 3 $[-0.0107, +0.0685]$ ID and $[-0.0124, +0.0378]$ OOD; TinyLlama $[-0.0016, +0.0066]$ ID and $[-0.0019, +0.0129]$ OOD. The largest point estimate is Gemma ID ($+0.0289$), but its interval is wide relative to the estimate. Teacher-matching KL is likewise close between learned and random cores on ID evaluation (0.2805/0.2877 for SmolLM2, 0.3003/0.3234 for Gemma, and 0.3082/0.3088 for TinyLlama). This post hoc uncertainty analysis does not establish equivalence; it shows that the accepted five-seed experiment does not resolve a stable advantage for the specific learned readout. [P7, P8]

This outcome follows from a structural identifiability problem in the tested boundary. Let the frozen scalar core be

$$s = w^T z + b,$$

where w is the learned 2,048-dimensional readout direction. If the final adapter map can be written locally as a trainable linear map $z = Ah(x)$, then

$$s = w^T Ah(x) + b = (A^T w)^T h(x) + b.$$

The effective readout direction in the target hidden space is therefore $A^T w$. For a nonzero fixed w , many choices of A can realize the same effective direction. In the implemented adapter, a nonlinear LayerNorm->Linear->GELU stack precedes the final trainable Linear(256,2048) projection, giving the adapter at least the freedom represented by this linear argument and additional upstream flexibility. The random control also preserves readout scale, bias, and temperature, preventing trivial differences in score magnitude or calibration from becoming a proxy for the identity of w . [P14]

Consequently, correct-teacher transfer can be genuine while specific-core portability remains unidentified. The adapter can learn an input-dependent mapping whose projection through either the learned or random scalar readout matches the teacher distribution. The success of that training says that the target backbone contains features from which the teacher’s behavior can be reconstructed through the adapter; it does not show that the pretrained Open-Jev readout supplies irreplaceable computation after the backbone change.

The correct interpretation is therefore not that decision 0006 was wrong or failed. It answered two different questions differently. It produced reproducible evidence for decision-behavior transfer, and at the same time identified why the particular linear core could not serve as a clean causal test of learned-core reuse. Decision 0007 was designed to remove this structural explanation rather than retrospectively reinterpret 0006 as a nonlinear-core test.

10. Decision 0006 external validation with JevBench

DecPort evaluated the Open-Jev teacher and one label-free-selected correct-teacher adapter per target using the pinned JevBench v1.4.0 harness at commit 2fa63fa3226cb369795525ed011800f57dcbd894. The adapter selection rule was the lowest final-epoch transfer-training loss and was applied before JevBench evaluation; JevBench answers were not used for tuning or selection. DecPort imported JevBench’s own public task loader, request builder, runner, scoring, and summary functions and verified the public tier file hashes. [P13, P14, E8]

The scope must be stated precisely. DecPort evaluated only the **JevBench v1.4.0 public subset: 231/534 tasks** - 72 original, 48 easy, and 111 hard. The remaining private/judge portion of the 534-task frozen set was not available to DecPort. In addition, JevBench v1.4.0’s official composite methodology incorporates a separate fresh sealed set of 308 decisions and blends it with prior axes; DecPort did not run or claim that official composite. [P13, E8]

Table E. Decision 0006 JevBench v1.4.0 public subset only (231/534 tasks)

System	Correct / 231	Choice / 139	Noul / 74	Score / 18	Brier	ECE
Open-Jev-2B teacher	151 (65.4%)	89	49	13	0.4752	0.140
DecPort + SmolLM2	92 (39.8%)	51	37	4	0.7474	0.166
DecPort + Gemma 3	81 (35.1%)	41	37	3	0.7042	0.125
DecPort + TinyLlama	78 (33.8%)	38	36	4	0.8556	0.309
Uniform guess (expected)	73.4 (31.8%)	—	—	—	—	—

Source: DecPort commit 51f2213; 0006 jevbench_public/summary.json plus system reports/provenance. Public subset only; not the full benchmark or sealed-inclusive composite.

On the 231 public tasks, the DecPort Open-Jev teacher replay scored 151/231 (65.4%). The accepted archive separately notes that Open-Jev’s project report published 150/231 for this released 2B checkpoint; the 151/231 value used here is the DecPort pinned replay, which also matched DecPort’s own smoke-fidelity check. SmolLM2 scored 92/231 (39.8%), Gemma 81/231 (35.1%), and TinyLlama 78/231 (33.8%), compared with an expected uniform-guess count of 73.4/231 (31.8%). Thus the ported targets are modestly above the uniform expectation but substantially below the teacher. The external result is consistent with partial behavior transfer rather than broad retention of teacher capability. [P7, P13, E6]

The type decomposition echoes the internal evaluation. Ported Noul scores are 36-37 correct out of 74 tasks, approximately chance, while the teacher reaches 49/74. Choice and Score also remain well below the teacher in absolute performance. These external measurements therefore do not rescue the Noul failure or imply that the target systems inherit Open-Jev’s full behavior. [P13]

TinyLlama requires a separate operational note. Its 2,048-token context limit caused 37 JevBench tasks to be refused rather than truncated: 28 Choice, 6 Noul, and 3 Score tasks. JevBench scored these as incorrect, so the primary result remains 78/231. TinyLlama was 78/194 on tasks it could execute, but that secondary denominator must not replace the benchmark denominator. [P7, P13, P14]

11. Final nonlinear ship gate (decision 0007): methods

Decision 0007 is a separate accepted experiment, not a replacement for decision 0006. It is the final preregistered test of the original RQ2 hypothesis after the 0006 linear-core confound was identified. The protocol was frozen at commit d0aa96f4defc3ce202996be1f75d669b791d57fc; the accepted result package and final report were archived at cf9939e213ca2c045bc6bf6968b68bcc8f8656be. [P20-P27]

11.1 Source representation boundary and nonlinear DecisionCore

The source representation remains the released ZefanCai/Open-Jev-2B system, with the same pinned source stack:

- model revision 0c7aa498b1627be8da4acf34c863ff0ee0a92785;
- Qwen/Qwen3.5-2B base revision 15852e8c16360a2fea060d615a32b45270f8a8fc;
- upstream Open-Jev commit 3308a15ccd7eea1df7a37d6ddc39b023b801ba16.

The Open-Jev backbone, rank-8 LoRA, original head, and original calibration remain frozen. A forward hook captures the exact float32 2,048-dimensional tensor entering the upstream Open-Jev head; reapplying that head to the captured representation is an integrity check. In the accepted archive the maximum absolute replay difference is 0.0. [P21, P24, P25]

The new learned DecisionCore is

$$2048 \rightarrow \text{LayerNorm}(2048) \rightarrow \text{Linear}(2048, 512) \rightarrow \text{GELU} \\ \rightarrow \text{Linear}(512, 128) \rightarrow \text{GELU} \rightarrow \text{Linear}(128, 1).$$

It has 1,118,977 float32 parameters, affine LayerNorm, and exact-erf GELU. Typed semantics are unchanged from 0006: Choice and Score score one representation per candidate; Noul scores one representation and assembles logits $[0, s]$ ordered (false,true). The learned core state SHA-256 in the accepted final gate is:

8d3ec3087e8cefe94f45562723b3bc851c523a245a1a3619c0f0b3104770000d. [P21-P25]

11.2 Source-only labeled core training, calibration, and leakage safeguards

The core was trained exactly once with labels on a new source-only split and then frozen for every seed, target, and learned-core condition. Source-core training used typed cross-entropy on uncalibrated typed logits (binary cross-entropy through the $[0, s]$ Noul construction), AdamW with learning rate 10^{-3} , weight decay 0.01, gradient clip 1.0, batch size 8 homogeneous in decision kind and width, and 10 epochs with no early stopping or model selection. A single calibration temperature was then fit on a separate 900-decision source calibration split by minimizing mean NLL over the fixed 4,001-point log-spaced grid from 0.05 to 20. The accepted temperature is 1.0842459532. [P21-P24]

The source-core records are deliberately disjoint from the 4,500 unlabeled target-transfer records and both evaluation splits. The 4,500 source-training decisions contain 1,500 each from ARC-Easy, BoolQ, and Yelp Review Full; calibration contains 300 each. ARC-Easy uses its otherwise unused test split, while BoolQ and Yelp use later positions from the same deterministic seed-0 shuffle. Data preparation found two exact ARC-Easy duplicates with transfer records; those shuffled indices (285 and 1063) were skipped before the full run, and the manifest verifies zero exact content overlap. [P21, P24]

The label boundary is explicit. Transfer-train answers are stripped immediately after loading; the target distillation path rejects labeled records; ID/OOD answers are read only for post-training evaluation; and the audit confirms zero answers in teacher or distillation inputs. These safeguards prevent source-core supervision from becoming target supervision through overlap or direct target labels. [P21, P25]

11.3 Frozen target backbones and deliberately weak rank-128 adapter

The frozen targets and exact resolved revisions are unchanged:

- SmolLM2-360M-Instruct: a10cc1512eabd3dde888204e902eca88bddb4951;
- Gemma 3 270M Instruct: 23cf460f6bb16954176b3ddcc8d4f250501458a9;
- TinyLlama-1.1B-Chat-v1.0: fe8a4ea1ffedaf415f4da2f062534de366a451e6.

Each target encodes Open-Jev candidate prompt text as plain text, with no chat template or truncation and a target maximum of 2,048 tokens. Last-token target states are cached once. [P24]

The 0006 $d \rightarrow 256 \rightarrow 2048$ nonlinear adapter is replaced by a deliberately constrained map:

$$\text{LayerNorm}(d) \rightarrow \text{Linear}(d, 128, \text{bias} = \text{false}) \rightarrow \text{Linear}(128, 2048, \text{bias} = \text{false}),$$

with no activation. Trainable parameter counts are 386,944 for SmolLM2 ($d = 960$), 345,344 for Gemma ($d = 640$), and 528,384 for TinyLlama ($d = 2048$). The purpose is not to make reparameterization mathematically impossible in every sense, but to remove the specific 0006 combination of a rank-one downstream readout and a nonlinear expressive adapter as the dominant explanation. [P21, P27]

11.4 Label-free transfer objective and conditions A-E

The target objective remains the 0006 calibrated decision-space objective:

$$L = \text{KL}(p_{\text{source}} \parallel p_{\text{target}}) + 0.1 \text{MSE}(\text{center}(\ell_{\text{target}}), \text{center}(\ell_{\text{source}})), \quad T = 1.$$

The teacher is now the source system *Open-Jev representation* $\rightarrow$ *learned nonlinear core* $\rightarrow$ *fitted calibration*. Detached outputs are computed once on the 4,500 unlabeled transfer inputs. Target training uses AdamW, learning rate 10^{-3} , weight decay 0.01, gradient clip 1.0, batch size 8, 10 epochs, and seeds 0-4. There is no target-label early stopping or selection. [P21-P24]

The five conditions are:

- **A - learned-core distillation.** Frozen learned nonlinear core; train only the rank-128 adapter against the correct source output.
- **B - random-core distillation.** Same adapter and teacher signal, but through a frozen random core with the same architecture and temperature.
- **C - mismatched-teacher distillation.** Learned core with source outputs deranged within decision kind and option count. The single five-option Choice record remains the one unavoidable fixed point (1/4,500 per seed).
- **D - untrained adapter.** The matched initialized adapter through the learned core, without adaptation.
- **E - target-specific distillation.** No shared core. A target-specific module $\text{LayerNorm}(d) \rightarrow \text{Linear}(d, 512) \rightarrow \text{GELU} \rightarrow \text{Linear}(512, 128) \rightarrow \text{GELU} \rightarrow \text{Linear}(128, 1)$ is trained directly on target states using the same unlabeled teacher outputs, typed assembly, calibration constant, objective, optimizer, epochs, batch size, and seed.

A-D share an identical adapter initialization per (seed, target). Target-specific parameter counts are 559,745 for SmolLM2, 395,265 for Gemma, and 1,118,977 for TinyLlama. [P21-P24, P27]

11.5 Scale-matched random nonlinear core

A random core must not lose merely because its untrained nonlinear layers emit a different numerical scale. Each seed therefore receives a separate random core with the learned core’s exact architecture and temperature. First, each parameter tensor is an independent standard-normal draw rescaled to the Frobenius norm of the corresponding learned tensor. Second, the Linear layers are matched sequentially on source-training representations: each receives a scalar gain on weight and bias

plus a scalar bias shift so that its pre-activation mean and standard deviation match the learned core at that layer. No learned direction is copied. [P21, P27]

This layer-wise scale matching was a preregistration correction made after a bounded smoke test and before the full run. Norm matching alone produced a raw candidate-score standard deviation of approximately 0.010 versus 3.55 for the learned core; with an input LayerNorm, that would have made the control a test of logit range rather than learned structure. The accepted record reports no protocol deviation and no restart after the full run began. [P20-P22]

11.6 Datasets, evaluation, and preregistered SHIP criteria

The transfer and evaluation splits are byte-identical to 0006: 4,500 transfer decisions (1,500 per type), 2,000 ID decisions (570 Choice, 700 Noul, 730 Score), and 1,500 OOD decisions (500 per type from OpenBookQA, QNLI, and Amazon Reviews Multi). Means and sample SDs are computed over five seeds. Metrics include accuracy, macro-F1, NLL, Brier score, 10-bin ECE, Score ordinal/expected-value MAE, Noul $P(\text{true})$ behavior, source top-choice agreement, KL, probability correlation, centered-logit MSE, parameter/storage cost, and runtime. [P23, P24]

JevBench uses v1.4.0 at commit 2fa63fa3226cb369795525ed011800f57dcdbd894, public subset only (231/534). For each trained target/condition, a seed is selected by the preregistered label-free rule: lowest final-epoch training loss, ties to the lowest seed; the untrained adapter uses seed 0. No JevBench answer is read for selection. Longer target tasks are refused and scored wrong. [P21, P27, E8]

The preregistered gate is mechanical:

- **Criterion A - learned shared core matters (primary).** A target requires mean A-B ID gain $\geq +0.05$, OOD gain $\geq +0.03$, and positive gain in at least 4/5 seeds on each split. The criterion requires at least 2/3 targets and no target with mean ID gain below -0.02.
- **Criterion B - input-specific transfer.** A target requires mean A-C ID gain $\geq +0.05$ and positive gain in at least 4/5 seeds; at least 2/3 targets must pass.
- **Criterion C - useful under shift.** On OOD, a target requires both A-C and A-D mean gains $\geq +0.03$, each positive in at least 4/5 seeds; at least 2/3 targets must pass.
- **Criterion D - external behavior.** A selected A adapter must exceed the JevBench uniform-guess expectation by at least 0.05×231 correct-equivalent margin; at least 2/3 targets must pass.
- **Criterion E - practical utility.** For each ID and OOD split, A must either beat E or come within 0.02 accuracy while using at most half as many per-target trainable parameters. Both splits must pass for a target, and at least 2/3 targets must pass. The fixed adapter/E parameter ratios are 0.691 (SmolLM2), 0.874 (Gemma), and 0.472 (TinyLlama).

SHIP requires $A \wedge B \wedge C \wedge D \wedge E$ plus a passing audit and reproduction; anything else is **NO-SHIP**. [P21-P22]

11.7 Audit and reproduction procedure

The final-gate audit checks data hashes and composition, source-core disjointness, Open-Jev head parity, unchanged source LoRA/head and target backbones, an unchanged and reloadable learned core, seed-specific random cores distinct from the learned core, matched A-D initialization, gradients only in allowed tensors, zero target labels in teacher/distillation inputs, the one expected mismatched fixed point, every-decision metric accounting, artifact hashes, and finite recorded values. The accepted archive also contains six tamper tests; altered adapter bytes, a changed core digest, an injected core gradient, a dropped Noul decision, a labeled distillation record, and a mismatched initialization were all rejected. [P25]

The full run was then re-executed from scratch with the identical command, code commit, config, data, and RTX 4060 Ti hardware environment. Acceptance required numerical agreement within 10^{-6} for compared result JSONs, identical recomputed internal criteria A, B, C, and E, and byte-identical safetensors. JevBench criterion D is the separate accepted public-subset evaluation and is not represented by the reproduction check’s criteria_a_b_c_e_identical field. This is a same-environment deterministic reproduction, not an independent-laboratory or cross-hardware replication. [P26]

12. Final ship-gate results (decision 0007)

12.1 Source system and calibration

The nonlinear source system is a competent teacher on the tested data. Open-Jev representations plus the learned core score 0.7455 ID and 0.6193 OOD overall, compared with 0.7310 ID and 0.6467 OOD for Open-Jev’s own linear head. The learned core’s source-training accuracy is 0.7964; held-out source-calibration accuracy is 0.7600 with NLL 0.5816; the fitted temperature is 1.0842459532. The source core is therefore not being evaluated merely as an untrained control, although its OOD accuracy is lower than the original Open-Jev head. [P23]

Table F. Decision 0007 source-system accuracy and calibration reference

System	Split	Overall	Choice	Noul	Score	NLL	ECE
Open-Jev repr. + nonlinear learned core	ID	0.7455	0.8930	0.8543	0.5260	0.5778	0.0227
Open-Jev repr. + nonlinear learned core	OOD	0.6193	0.6320	0.6920	0.5340	0.9282	0.0945
Open-Jev 2B own linear head	ID	0.7310	0.8877	0.7929	0.5493	0.6912	0.0808
Open-Jev 2B own linear head	OOD	0.6467	0.6080	0.8600	0.4720	0.8465	0.0747

Source: Decision 0007 accepted archive `run/report_tables.md` and `run/source_system/summary.json`.

12.2 Overall target results and the primary learned-core test

Table G reports all five 0007 conditions. The central RQ2 comparison is A versus B under the same correct source signal. The learned nonlinear core does not produce the preregistered accuracy advantage over the scale-matched random nonlinear core on any target. SmolLM2 A-B is effectively zero ID (-0.0000 ± 0.0289 ; positive in 1/5 seeds) and $+0.0057 \pm 0.0070$ OOD (5/5). Gemma is $+0.0147 \pm 0.0394$ ID (2/5) and $+0.0001 \pm 0.0098$ OOD (2/5). TinyLlama is -0.0145 ± 0.0216 ID (2/5) and -0.0113 ± 0.0065 OOD (0/5). None approaches the preregistered $+0.05$ ID and $+0.03$ OOD mean-gain requirements simultaneously. [P22-P23]

Table G. Decision 0007 overall accuracy, mean $\pm$ sample SD over five seeds

Target	Split	A learned	B random	C mismatch	D untrained	E target-specific
SmolLM2	ID	0.5137 $\pm$.0061	0.5137 $\pm$.0271	0.3696 $\pm$.0109	0.3635 $\pm$.0126	0.5192 $\pm$.0120
SmolLM2	OOD	0.3564 $\pm$.0080	0.3507 $\pm$.0114	0.3141 $\pm$.0118	0.3137 $\pm$.0115	0.3523 $\pm$.0104
Gemma 3	ID	0.4501 $\pm$.0091	0.4354 $\pm$.0411	0.3453 $\pm$.0068	0.3195 $\pm$.0489	0.4565 $\pm$.0188
Gemma 3	OOD	0.3507 $\pm$.0172	0.3505 $\pm$.0162	0.3100 $\pm$.0139	0.3100 $\pm$.0132	0.3671 $\pm$.0223
TinyLlama	ID	0.5172 $\pm$.0126	0.5317 $\pm$.0161	0.3482 $\pm$.0179	0.3283 $\pm$.0303	0.5347 $\pm$.0060
TinyLlama	OOD	0.3783 $\pm$.0063	0.3896 $\pm$.0097	0.3053 $\pm$.0100	0.3116 $\pm$.0206	0.3824 $\pm$.0046

Source: Decision 0007 accepted archive `run/report_tables.md`. $n = 5$ seeds; SD is sample SD. ID $N = 2,000/\text{seed}$; OOD $N = 1,500/\text{seed}$.

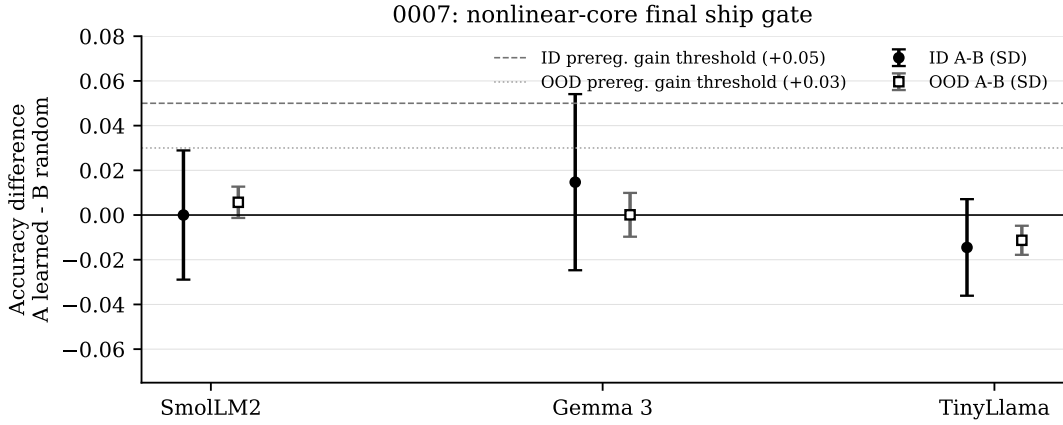


Figure 3: Decision 0007 learned-core minus random-core accuracy effect. Error bars are sample SD across five paired seeds. Horizontal lines show the preregistered mean-gain thresholds; criterion A also requires at least 4/5 positive seeds on each split.

The learned core does fit the source distributions more steadily in distribution: ID KL is 0.4929/0.5437/0.5250 for SmolLM2/Gemma/TinyLlama versus 0.9209/0.6329/0.5944 for the random core, and ID ECE is 0.0507/0.0520/0.0689 versus 0.1228/0.1156/0.1026. Those fidelity and calibration differences do not become an accuracy advantage under the preregistered criterion. The negative RQ2 result is therefore not a claim that every metric is identical; it is that the learned shared core fails to demonstrate the required reusable decision advantage. [P23]

12.3 Input-specific transfer and OOD behavior remain positive

Behavior transfer remains separable from core portability. Learned-core A versus mismatched-teacher C gains are $+0.1441 \pm 0.0082$, $+0.1048 \pm 0.0133$, and $+0.1690 \pm 0.0095$ ID for SmolLM2, Gemma, and TinyLlama respectively, positive in 5/5 seeds for every target. The corresponding OOD gains shrink to $+0.0423 \pm 0.0190$, $+0.0407 \pm 0.0139$, and $+0.0729 \pm 0.0122$, but remain positive in 5/5 seeds. A also exceeds the untrained D condition OOD by $+0.0427$, $+0.0407$, and $+0.0667$ on the three targets, satisfying criterion C on all three. Criteria B and C therefore pass even though criterion A fails. [P22-P23]

Table H. Decision 0007 decisive paired accuracy differences

Target	Split	A-B learned-random	A-C correct-mismatch	A-E learned-target-specific
SmolLM2	ID	$-0.0000 \pm .0289$ (1/5)	$+0.1441 \pm .0082$ (5/5)	$-0.0055 \pm .0109$
SmolLM2	OOD	$+0.0057 \pm .0070$ (5/5)	$+0.0423 \pm .0190$ (5/5)	$+0.0041 \pm .0117$
Gemma 3	ID	$+0.0147 \pm .0394$ (2/5)	$+0.1048 \pm .0133$ (5/5)	$-0.0064 \pm .0248$
Gemma 3	OOD	$+0.0001 \pm .0098$ (2/5)	$+0.0407 \pm .0139$ (5/5)	$-0.0164 \pm .0299$
TinyLlama	ID	$-0.0145 \pm .0216$ (2/5)	$+0.1690 \pm .0095$ (5/5)	$-0.0175 \pm .0138$
TinyLlama	OOD	$-0.0113 \pm .0065$ (0/5)	$+0.0729 \pm .0122$ (5/5)	$-0.0041 \pm .0061$

Source: Paired per-seed differences from the accepted 0007 `run/report_tables.md`. Parentheses in A-B and A-C are seeds with positive paired gain.

12.4 Practical value versus target-specific distillation

The target-specific baseline E uses the same unlabeled source outputs, objective, optimizer, epoch count, batch size, and per-target training stage but does not reuse a shared core. It matches or exceeds A in mean accuracy on every listed split except SmolLM2 OOD, where A is higher by 0.0041. A-E is -0.0055 ID/+0.0041 OOD for SmolLM2, -0.0064/-0.0164 for Gemma, and -0.0175/-0.0041 for TinyLlama. The per-target adapter/E parameter ratios are 0.691, 0.874, and 0.472. Under the frozen criterion E, only TinyLlama passes via the “within 2 pp and at most half the parameters” route; SmolLM2 and Gemma do not. Criterion E therefore fails overall. The shared core additionally contributes 1,118,977 parameters stored once and requires the source-side training/calibration stage. [P22-P23]

12.5 Decision types, Noul failure, and calibration

Choice and Score carry most of the positive behavior result. Under learned-core A, ID Choice accuracy is 0.6561 ± 0.0134 on SmolLM2, 0.4568 ± 0.0145 on Gemma, and 0.5263 ± 0.0126 on TinyLlama, while the mismatched control is 0.2814, 0.2133, and 0.2558. Score similarly exceeds mismatch on all targets ID and OOD. Choice OOD is weaker: Gemma A is 0.2648 ± 0.0171 versus mismatch 0.2692 ± 0.0397 , while SmolLM2 and TinyLlama retain small positive mean differences. [P23]

Noul fails again, but with a different marginal collapse than 0006. ID Noul accuracy under A is 0.6366 ± 0.0113 on SmolLM2, 0.6291 ± 0.0296 on Gemma, and 0.6011 ± 0.0425 on TinyLlama, no better than the corresponding mismatched means 0.6386, 0.6386, and 0.6186. Learned-core students predict true on 93.3%, 92.4%, and 61.7% of ID Noul decisions respectively. OOD predicted-true rates rise to 99.2%, 86.7%, and 99.6%, while OOD Noul accuracy remains within about 0.02 of mismatch. The source system itself is 0.8543 ID and 0.6920 OOD on Noul. The positive aggregate behavior-transfer conclusion therefore does not extend uniformly to Noul. [P23]

Table I. Decision 0007 learned-core condition by type, compared with mismatch and source

Target	Split	Type	Source	A learned core	C mismatch
SmolLM2	ID	Choice	0.8930	$0.6561 \pm .0134$	$0.2814 \pm .0295$
SmolLM2	ID	Noul	0.8543	$0.6366 \pm .0113$	$0.6386 \pm .0000$
SmolLM2	ID	Score	0.5260	$0.2847 \pm .0114$	$0.1805 \pm .0214$
SmolLM2	OOD	Choice	0.6320	$0.3324 \pm .0148$	$0.2644 \pm .0186$
SmolLM2	OOD	Noul	0.6920	$0.4736 \pm .0026$	$0.4760 \pm .0000$
SmolLM2	OOD	Score	0.5340	$0.2632 \pm .0146$	$0.2020 \pm .0171$
Gemma 3	ID	Choice	0.8930	$0.4568 \pm .0145$	$0.2133 \pm .0183$
Gemma 3	ID	Noul	0.8543	$0.6291 \pm .0296$	$0.6386 \pm .0000$
Gemma 3	ID	Score	0.5260	$0.2732 \pm .0323$	$0.1671 \pm .0064$
Gemma 3	OOD	Choice	0.6320	$0.2648 \pm .0171$	$0.2692 \pm .0397$
Gemma 3	OOD	Noul	0.6920	$0.4908 \pm .0202$	$0.4760 \pm .0000$
Gemma 3	OOD	Score	0.5340	$0.2964 \pm .0236$	$0.1848 \pm .0181$
TinyLlama	ID	Choice	0.8930	$0.5263 \pm .0126$	$0.2558 \pm .0356$
TinyLlama	ID	Noul	0.8543	$0.6011 \pm .0425$	$0.6186 \pm .0440$
TinyLlama	ID	Score	0.5260	$0.4296 \pm .0155$	$0.1611 \pm .0160$
TinyLlama	OOD	Choice	0.6320	$0.2932 \pm .0106$	$0.2396 \pm .0242$
TinyLlama	OOD	Noul	0.6920	$0.4768 \pm .0018$	$0.4900 \pm .0183$
TinyLlama	OOD	Score	0.5340	$0.3648 \pm .0119$	$0.1864 \pm .0130$

Source: Decision 0007 accepted archive `run/report_tables.md`; means $\pm$ sample SD over five seeds.

OOD calibration remains weak. Learned-core A OOD ECE is 0.2186 on SmolLM2, 0.1451 on Gemma, and 0.3130 on TinyLlama. TinyLlama’s OOD NLL is 3.5446 ± 1.3515 , compared with 1.3903 ± 0.1363 for its untrained adapter. The final-gate evidence therefore supports smaller OOD accuracy effects, not robust OOD calibration. [P23]

12.6 JevBench public subset and mechanical verdict

The 0007 source system scores 137/231 on the JevBench public subset, lower than the 151/231 replay of Open-Jev’s own linear head in decision 0006. This is expected to be treated as a distinct source system, not as the same teacher result. Selected learned-core A adapters score 89/231 on SmolLM2, 73/231 on Gemma, and 74/231 on TinyLlama. The matched random-core selected systems score 87, 73, and 74 respectively. The target-specific modules score 96, 70, and 65. Uniform guessing expects 73.37 correct. Only SmolLM2’s A system clears the preregistered uniform-plus-5-percentage-point threshold; criterion D therefore fails. TinyLlama again refuses 37 tasks beyond its 2,048-token context, and those tasks remain scored wrong under the 231-task denominator. [P27]

Table J. Decision 0007 JevBench public subset (231/534) and final criteria

System	SmolLM2	Gemma	TinyLlama
A learned core	89	73	74
B random core	87	73	74
C mismatched	85	78	64
D untrained	69	73	83
E target-specific	96	70	65
Source system	137 / 231		
Uniform expected	73.37 / 231		

Criterion	Pass	Result
A learned core matters	none	FAIL
B input-specific	3/3	pass
C useful under shift	3/3	pass
D external behavior	SmolLM2	FAIL
E practical utility	TinyLlama	FAIL
Audit	–	pass
Reproduction	–	pass
Final verdict	NO-SHIP	

Source: Decision 0007 jevbench_public/summary.json, verdict.json, and final report. JevBench is public subset only; TinyLlama A has 37 context refusals scored wrong.

Because SHIP required all five criteria together with the audit and reproduction, failures of A, D, and E mechanically produce **NO-SHIP**. The primary RQ2 criterion A fails on all three targets. The result does not imply that “nothing transferred”: criteria B and C pass on all three targets, directly preserving the separate behavior-transfer finding. [P22-P23]

12.7 Final-gate audit and bit-exact reproduction

The accepted audit reports a passing status with Open-Jev head parity maximum absolute difference 0.0, source-system output recomputation difference 0.0, the learned core digest unchanged and reloadable, seed-specific random cores distinct from the learned core, unchanged target backbone digests, matched A-D initialization, gradients only in the allowed adapter/module tensors, source-core disjointness, strict label exclusion, every decision counted, and artifact hashes verified. Six deliberately tampered copies were rejected as described in Section 11.7. [P25]

The reproduction comparison covers 93 result JSON files with maximum absolute numeric difference 0.0 and 81/81 safetensors artifacts byte-identical; recomputed internal criteria A, B, C, and E are identical, and the reproduction’s own audit passed. Both executions used the same code/config/data and the same RTX 4060 Ti hardware/software environment; this establishes deterministic same-environment repeatability, not independent replication. [P26]

A provenance nuance is retained explicitly: both the accepted run and reproduction record the protocol commit d0aa96f4defc3ce202996be1f75d669b791d57fc while also recording git_worktree_dirty:true. The archive nevertheless verifies the frozen repository configuration and data hashes, and the second execution is bit-exact. Accordingly, the commit is reported as the recorded base/protocol revision, but this article does not describe the run as originating from a clean worktree. [P24, P26]

13. Limitations

The final evidence directly addresses the principal structural confound identified in decision 0006, but it remains bounded. The limitations should therefore distinguish the limitations that motivated decision 0007 from those that remain after it.

- **Decision 0006 linear-core and adapter-capacity limitations were addressed, not ignored.** The rank-one Linear(2048->1) core and nonlinear $d \rightarrow 256 \rightarrow 2048$ adapter explain why 0006 could not identify learned-core reuse. Decision 0007 replaced them with a nonlinear 1.12M-parameter core and a rank-128 linear adapter. Those are limitations of the 0006 RQ2 test, not the final evidence.
- **Five seeds.** Both accepted stages use five target-training seeds. Decision 0007 uses preregistered effect thresholds and positive-seed counts rather than a post hoc equivalence claim. Failure of criterion A means the required advantage was not demonstrated; it does not prove exact equivalence of learned and random cores.
- **Three target families and one source representation system.** The final gate covers SmolLM2-360M-Instruct, Gemma 3 270M Instruct, TinyLlama-1.1B-Chat, and one Open-Jev/Qwen source representation boundary. It should not be extrapolated to arbitrary architectures, scales, or representation boundaries.
- **One nonlinear core design and finite training/data budget.** Decision 0007 tests one compact MLP DecisionCore, one rank-128 adapter family, 4,500 source-core training decisions, 900 calibration decisions, 4,500 unlabeled transfer decisions, ten epochs, and the recorded optimization settings. Other boundaries or substantially different architectures remain untested.
- **Public JevBench subset only.** Both accepted stages evaluate only the 231/534 public frozen tasks, not the private/judge portion or JevBench v1.4.0’s sealed-inclusive official composite.
- **Same-environment reproduction.** The accepted runs reproduce bit-exactly in the same RTX 4060 Ti/software setting. There is no cross-hardware replication and no independent research-group replication.
- **OOD behavior and calibration remain weaker.** Input-specific accuracy gains shrink OOD, and final-gate OOD calibration remains poor, including TinyLlama A OOD NLL 3.54 ± 1.35 .
- **Persistent Noul failure.** Noul does not recover input-specific source behavior under either accepted stage. Decision 0006 collapses mainly toward false on two targets; decision 0007 produces a different near-constant/high-true failure. Choice and Score carry the positive result.
- **Prompting/context asymmetries.** Teacher and targets share Open-Jev candidate rendering but tokenize and encode it with different frozen backbones. Internal target context is capped at 2,048 while the source allows 4,096; TinyLlama refuses 37 JevBench tasks under this policy.

- **Provenance worktree flag.** Decision 0007 provenance records the frozen protocol commit but also `git_worktree_dirty: true`. The accepted archive and reproduction establish exact result repeatability under that recorded state, but the provenance should not be described as a clean-checkout guarantee.

14. Interpretation

14.1 Supported: label-free input-specific behavior transfer

RQ1 is supported within the tested setting. Across both accepted Open-Jev stages, lightweight model-specific adaptation transfers input-specific probabilistic decision behavior from a real source system into three heterogeneous frozen backbones without target ground-truth labels. In decision 0006, correct teacher exceeds mismatch overall in all five seeds on every target. In the stricter decision 0007 gate, A-C ID gains are +0.144, +0.105, and +0.169 for SmolLM2, Gemma, and TinyLlama, with smaller but positive OOD gains +0.042, +0.041, and +0.073; preregistered criteria B and C pass on all three targets. Choice and Score carry most of this result, while Noul remains problematic. [P7-P8, P22-P23]

The supported claim is therefore *behavior distillation*, not learned-core portability. A target can reconstruct a useful portion of the source system’s input-specific decision function from frozen-backbone features through a small trained interface. The target systems do not reproduce the source completely, and OOD/calibration results are weaker.

14.2 Not supported: reusable learned DecisionCore portability

RQ2 is not supported by the accepted evidence. Decision 0006 could not identify learned-core reuse because a scalar linear readout can be absorbed into the expressive adapter’s trainable projection. Decision 0007 directly addresses that principal structural explanation and still finds no target satisfying the preregistered learned-core criterion. Learned nonlinear core versus scale-matched random nonlinear core mean accuracy differences remain near zero or negative relative to the +0.05 ID/+0.03 OOD gate, and the target-specific distilled baseline matches or exceeds the learned-core route at comparable per-target scale under the preregistered utility test. Criterion A fails on all targets and criterion E passes only TinyLlama, producing NO-SHIP when combined with the external criterion D failure. [P22-P23]

This is not evidence that nothing transferred. Behavior transferred. The evidence does not show that the reusable learned shared core is what carried that transfer. Under the tested models, data, architecture, boundary, and training budget, a learned shared DecisionCore provides no demonstrated practical accuracy advantage over the matched random core or target-specific distillation sufficient to satisfy the frozen criteria.

14.3 Open questions after decision 0007

The nonlinear-core experiment is no longer an unrun requirement; it is completed evidence. The remaining questions are narrower and genuinely downstream of the final gate:

- Can a materially different core boundary or substantially different source/core architecture yield learned structure that remains useful across backbone changes when the adapter is constrained?
- Do the same conclusions hold across larger and more diverse source/target model families, scales, and representation systems?
- Can cross-backbone behavior distillation retain a stronger OOD effect and materially better calibration without access to target labels?
- Why does Noul fail under both accepted designs, despite different marginal-collapse directions, and what representation or objective change is required to recover input-specific Boolean behavior?
- Does the archived result reproduce across hardware/software stacks and by an independent implementation or research group?

These are future experimental questions rather than claims about outcomes not present in the repository.

Data and code availability

All DecPort evidence cited in this article is archived in the public repository <https://github.com/suffro/decport>. Provenance is stage-specific and should not be collapsed into one commit. Decision 0006’s accepted archive is present at evidence commit 51f221348163b320e46111d0857a4a255e304ec0; its experiment provenance records implementation commit 0ab715530a3facf5d95cba9bedf6949f82ccdd5e. Decision 0007’s protocol and implementation were frozen at d0aa96f4defc3ce202996be1f75d669b791d57fc; the final accepted 0007 results, report, audit, reproduction package, and current repository state inspected for this article are archived at cf9939e213ca2c045bc6bf6968b68bcc8f8656be. [P7-P14, P20-P27]

The 0007 archive includes SHA-256 data/config/artifact checks, exact model and code revisions, per-seed results, aggregate tables, verdict.json, six audit failure-path probes, and a bit-exact second execution. JevBench per-item public task content is not copied into DecPort; aggregate reports, provenance, selections, and hashes of excluded raw artifacts are retained. [P22-P27]

Evidence note

All quantitative DecPort values in this article are transcribed from or computed directly from the repository’s archived JSON/Markdown result artifacts. Diagnostic measurements are labeled as diagnostics and are not combined with accepted benchmark evidence. Decision 0006 values remain associated with its linear Open-Jev core experiment; decision 0007 values remain associated with the nonlinear final gate. Internal accepted standard deviations are the repository’s sample SD ($n - 1$) across seeds 0-4. JevBench values are single label-free-selected system runs on the public subset and therefore are not reported with seed SD. The 0006 post hoc confidence intervals are descriptive and are not treated as equivalence tests. The 0007 final verdict is the repository’s preregistered mechanical gate, not a retrospective score.

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